Bernoulli's and Laplace Equation

Important assumptions in fluid dynamics:

- ➔ Inviscid, $\mu = 0$ (viscosity is approximately equal to zero)
- ➔ Incompressible, ρ = constant
- ➔ Irrotational, $\xi = 0$

These assumptions lead to the development of a powerful governing equation developed by Bernoulli and the powerful tools by Laplace in the context of aerodynamics.

Bernoulli's Equation:

$$p + \frac{1}{2}\rho U^2 = \text{constant}$$

Laplace Equation:

$$\nabla^2 f = 0$$ (**Second-order linear partial differential equation**)

If the flow is inviscid, it need not be irrotational.

$$\text{Inviscid} \neq \text{Irrotational}$$

To produce rotation of the fluid element, shear stress is required.

Consider the conservation of momentum in the x-direction for an incompressible flow,

$$\rho\frac{Du}{Dt} = \boxed{-\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) + \rho a_x}$$

Terms that produce acceleration

$$\frac{\partial p}{\partial x} + \mu\frac{\partial^2 u}{\partial x^2} = \text{Terms associated with normal stress}$$

$$\frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \text{Terms associated with shear stress}$$

If the flow is inviscid ($\mu = 0$), all the shear-stress-related terms go away. Nothing will cause rotation.

When to assume the flow to be inviscid:

1. Away from the boundaries (outside the boundary layer)
2. High Reynolds number (inertial force becomes more important than the viscous force and the viscous force can eventually be neglected)

3. Low-velocity gradient

Incompressible:

1. Low speed, Mach number < 0.3 (Good rule of thumb)
2. Liquids

Bernoulli's Equation: States that for an inviscid and incompressible fluid along a streamline

$$p + \frac{1}{2}\rho U^2 = \text{constant}$$

Main assumptions for Bernoulli's equation:

- ➔ Inviscid, $\mu = 0$
- ➔ Incompressible, ρ = constant
- ➔ No body forces, $\rho a_x = 0$
- ➔ Steady, $\partial(\ )/\partial t = 0$
- ➔ Along streamline

Bernoulli's equation can be derived from the momentum equation along x, y, and z. It is expressed as

$$\left.\begin{aligned}
&\mathbf{x-direction} \rightarrow \rho\left[u\frac{\partial \mathbf{u}}{\partial x} + v\frac{\partial \mathbf{u}}{\partial y} + w\frac{\partial \mathbf{u}}{\partial z}\right] = -\frac{\partial P}{\partial \mathbf{x}} \\
&\mathbf{y-direction} \rightarrow \rho\left[u\frac{\partial \mathbf{v}}{\partial x} + v\frac{\partial \mathbf{v}}{\partial y} + w\frac{\partial \mathbf{v}}{\partial z}\right] = -\frac{\partial P}{\partial \mathbf{y}} \\
&\mathbf{z-direction} \rightarrow \rho\left[u\frac{\partial \mathbf{w}}{\partial x} + v\frac{\partial \mathbf{w}}{\partial y} + w\frac{\partial \mathbf{w}}{\partial z}\right] = -\frac{\partial P}{\partial \mathbf{z}}
\end{aligned}\right\}$$

Convective acceleration (L.H.S) is due to pressure gradient in a particular direction

These are the equations in cartesian space.

However, Bernoulli's equation is derived along a streamline. Consider a fluid parcel moving through the space, it follows the streamline.

Hence the position of a fluid element can be defined by a different function of the position along the curve s, instead of x, y, and z.

Position = f(s)

Also, we can consider the velocity as a single parameter, because, by the definition the velocity is always tangent to the curve (i.e. streamline). So, there is always a single total velocity component (U).

u, v, w → U

The three equations from the conservation of momentum in cartesian space can be converted into a single equation in the streamline space by considering it along a streamline.

$$\rho\left[\mathrm{u}\frac{\partial \mathbf{u}}{\partial \mathrm{x}}+\mathrm{v}\frac{\partial \mathbf{u}}{\partial \mathrm{y}}+\mathrm{w}\frac{\partial \mathbf{u}}{\partial \mathrm{z}}\right]=-\frac{\partial \mathrm{P}}{\partial \mathbf{x}}$$

Multiply by dx on both the sides, we get **Equation – 1**

$$\rho\left[\mathrm{u}\frac{\partial \mathbf{u}}{\partial \mathrm{x}}\mathrm{dx}+\mathrm{v}\frac{\partial \mathbf{u}}{\partial \mathrm{y}}\mathrm{dx}+\mathrm{w}\frac{\partial \mathbf{u}}{\partial \mathrm{z}}\mathrm{dx}\right]=-\frac{\partial \mathrm{P}}{\partial \mathbf{x}}\mathrm{dx}$$

Along a streamline, **Equation – 2**

$$\mathrm{udz}-\mathrm{wdx}=0\Rightarrow \mathrm{dx}=\frac{\mathrm{u}}{\mathrm{w}}\mathrm{dz}$$

$$\mathrm{vdx}-\mathrm{udy}=0\Rightarrow \mathrm{dx}=\frac{\mathrm{u}}{\mathrm{v}}\mathrm{dy}$$

Plug the Equation – 2 in Equation – 1, we get

$$\mathrm{u}\frac{\partial \mathbf{u}}{\partial \mathrm{x}}\mathrm{dx}+\mathrm{u}\frac{\partial \mathbf{u}}{\partial \mathrm{y}}\mathrm{dy}+\mathrm{u}\frac{\partial \mathbf{u}}{\partial \mathrm{z}}\mathrm{dz}=-\frac{1}{\rho}\frac{\partial \mathrm{P}}{\partial \mathbf{x}}\mathrm{dx}$$

From calculus, if u = f(x, y, z) then **Equation – 3**

$$\mathrm{du}=\frac{\partial \mathrm{u}}{\partial \mathrm{x}}\mathrm{dx}+\frac{\partial \mathrm{u}}{\partial \mathrm{y}}\mathrm{dy}+\frac{\partial \mathrm{u}}{\partial \mathrm{z}}\mathrm{dz}$$

So, Equation – 3 becomes,

$$\mathrm{udu}=-\frac{1}{\rho}\frac{\partial \mathrm{p}}{\partial \mathrm{x}}\mathrm{dx}$$

Equation – 3 can also be written as,

$$\frac{1}{2}\mathrm{d}\left(\mathrm{u}^{2}\right)=-\frac{1}{\rho}\frac{\partial \mathrm{p}}{\partial \mathrm{x}}\mathrm{dx}$$

Similarly, for y- and z – directions

$$\frac{1}{2}\mathrm{d}\left(\mathrm{v}^{2}\right)=-\frac{1}{\rho}\frac{\partial \mathrm{p}}{\partial \mathrm{y}}\mathrm{dy}\Rightarrow \mathrm{y}-\mathrm{direction}$$

$$\frac{1}{2}\mathrm{d}\left(\mathrm{w}^{2}\right)=-\frac{1}{\rho}\frac{\partial \mathrm{p}}{\partial \mathrm{z}}\mathrm{dz}\Rightarrow \mathrm{z}-\mathrm{direction}$$

Add up all the three relations, we get

$$\frac{1}{2}d\left(u^2+v^2+w^2\right)=-\frac{1}{\rho}\left(\frac{\partial p}{\partial x}dx+\frac{\partial p}{\partial y}dy+\frac{\partial p}{\partial z}dz\right)$$

$$\frac{1}{2}d\left(U^2\right)=-\frac{1}{\rho}dp \Rightarrow UdU=-\frac{1}{\rho}dp$$

Along a streamline,

$$\rho UdU=-dp$$

Integrating on both the sides,

$$\rho\int UdU=-\int dp$$

$$\frac{1}{2}\rho U^2=-p+\text{constant}$$

Bernoulli's equation:

$$p+\frac{1}{2}\rho U^2=\text{constant}$$

If the flow is rotational, $\xi \neq 0$

Bernoulli's equation is valid along a streamline

If the flow is irrotational, $\xi = 0$

Example: Flow over an airfoil

Bernoulli's equation is valid everywhere in the flow field.

Example: Slow flow through converging duct.

Laplace Equation:

$\nabla^2 f=0$ (**Second-order linear partial differential equation**)

- **The most famous equation in mathematical physics**

If the Laplacian of a function equals zero, it is said to satisfy the Laplace equation.

Laplace equation can be used for:

- ➔ Electromagnetism
- ➔ Heat
- ➔ Fluids

For fluids, f = Ψ (stream function) or ϕ (velocity potential)

These two functions satisfy the Laplace equation if we make the following assumptions:

- **Inviscid**
- **Incompressible**
- **Irrotational**

Laplace equation for velocity potential:

The definition of velocity potential comes from equating the velocity component to their corresponding spatial gradients of the potential.

$$u = \frac{\partial \phi}{\partial x};\ v = \frac{\partial \phi}{\partial y};\ w = \frac{\partial \phi}{\partial z}$$

Since the flow is incompressible,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Substituting u, v, and w in the second equation, we get

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = 0 = \nabla^2 \phi$$

(Differential form of the Laplacian operator equating to zero)

The velocity potential due to conservation of mass satisfies the Laplace equation.

For cylindrical coordinates,

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

For spherical coordinates,

$$\frac{1}{r^2 \sin\theta}\left[\frac{\partial}{\partial r}\left(r^2 \sin\theta \frac{\partial \phi}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right) + \frac{\partial}{\partial \Phi}\left(\frac{1}{\sin\theta}\frac{\partial \phi}{\partial \Phi}\right)\right] = 0$$

Laplace equation for stream function:

From the definition of stream function,

$$u = \frac{\partial \psi}{\partial y};\ v = -\frac{\partial \psi}{\partial x}$$

Since the flow is irrotational (ξ = 0),

$$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$ **(Irrotational condition for 2D flow)**

Substituting u and v into the above equation, we get

$$\frac{\partial}{\partial x}\left(-\frac{\partial \psi}{\partial x}\right) - \frac{\partial}{\partial y}\left(\frac{\partial \psi}{\partial y}\right) = 0 \Rightarrow \frac{\partial^2 \psi}{\partial x^2} - \frac{\partial^2 \psi}{\partial y^2} = 0 = \nabla^2 \psi$$

- Any irrotational, incompressible flow has a velocity potential and stream function (for two-dimensional flow) that both satisfy Laplace's equation.

- Conversely, any solution of Laplace's equation represents the velocity potential or stream function (two-dimensional) for an irrotational, incompressible flow.

For Laplace equation,

- ➔ Solution already exists
- ➔ Solutions of the Laplace equation are called **Harmonic functions**.
- ➔ Useful in theoretical/numerical aerodynamics.
- ➔ Many CFD solvers solve the Laplace equation away from the boundaries, which is much easier than solving the conservation equations completely.
- ➔ Near the boundaries, the Laplace equation cannot be used (**flow is viscous**).

In practice, Bernoulli's equation is used a lot.

Note that Laplace's equation is a second-order linear partial differential equation. The fact that it is *linear* is particularly important because the sum of any particular solutions of a linear differential equation is also a solution of the equation (**Superposition principle**).

- Consider the irrotational, incompressible flow fields over different aerodynamic shapes, such as a sphere, cone, or airplane wing.

- These different flows are all governed by the same equation,

$$\nabla^2 \phi = 0$$

- How, then, do we obtain different flows for the different bodies? **The answer is found in the *boundary conditions.***

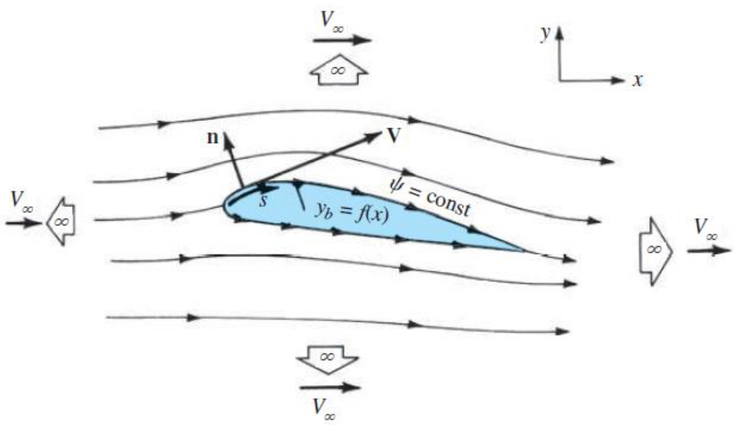
- Boundary conditions are therefore of vital concern in aerodynamic analysis.

Two sets of boundary conditions apply as follows:

1. Infinity boundary condition
2. Wall boundary condition

Infinity boundary condition:

- Far away from the body in all the directions, the flow approaches uniform freestream condition.
- Let V_∞ be aligned with x direction as shown in Figure.



Boundary conditions at infinity and on a body; inviscid flow.

$$u = \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = V_\infty$$

$$v = \frac{\partial \phi}{\partial y} = \frac{\partial \psi}{\partial x} = 0$$

Wall boundary condition:

- The flow cannot penetrate the surface, the velocity vector must be tangent to the surface.
- If the flow is tangent to the surface, then the component of velocity *normal* to the surface must be zero.
- The wall boundary condition can be written as

$$\vec{V} \bullet \vec{n} = (\nabla \phi) \bullet \vec{n} = 0$$

$$\frac{\partial \phi}{\partial n} = 0$$

- If we are dealing with Ψ rather than ϕ, then the wall boundary condition is

$$\frac{\partial \psi}{\partial s} = 0$$

Reference:

Anderson, J. (2011)*: Fundamentals of Aerodynamics (SI units)*. McGraw hill.